Sprint Progress Report 5

Sprint Overview

Development of Website and Construction of Database.

In Sprint 5, we focused on closing the loop between authentication and group workflow and the core expense APIs – so we can move into settlement, reminders and admin support in Sprint 6. The goal of this sprint was to convert existing, albeit disconnected, backend layers and front-end pages into a more connected MVP flow. This means login state was preserved across pages; group membership was handled consistently in the service layer; and expense history data was exposed through reusable API responses.

The codebase has a clearer separation of frontend state management and backend business logic. The authentication endpoints are used by the login and register pages, the stored user information is used to prefill the payer field for the expense form and the dashboard/history pages consume normalized expense data instead of static mock content. At the backend, repository and service layer already facilitated group creation, automatic creator membership, expense split, settlement record and activity logging, obviously giving us a stable footing for the remaining sprint work.

1. Class Presentation

exhibition to peers.

The check-in process was maintained on a weekly basis to continue reviewing the actual state of implementation and validating that what is remaining stays within MVP scope.

2. Work Completed

Work Completed Essay

Login Stage in Frontend - The pages for logging in and registering link to /api/login and /api/register, allowing you to create a user account, and authenticate against the backend. - The returned user payload is stored in `localStorage` by the login page, which gives a session-like state to application across page navigation. - The Add Expense page checks if the user is signed in before submitting and automatically fills the payer email from the user record in the DB. - The information in the Dashboard and History pages is live data from the expenses. Therefore, the account state that you see is not hard-coded in the UI but is coming from the backend.

A group workflow and a flow of the main expense. - The group is being created in the back-end repository layer where user responsible for the creation is set as ADMIN by default. Member is also being added to the group through join table. - The expense service utilizes the get_or_create_group_by_name method to facilitate the creation of a shared group. This maintains consistent group flow throughout the MVP journey. - The recording of an expense now only records the main expense and participants row if it passes relevant validations. - The responses sent back to the front end for expense history contain the group name, payer name, date, participants, and settlement flags. This is enough structure for the frontend to render the ledger and dashboard summaries. - The logic for the settlement repository is in place. Repayment transactions are recorded with activity log entries, so the app is ready for the settlement work in the next sprint.

Coverage of API and validation. - The new API surface contains login, register, expense history, and expense creation endpoints used by frontend pages. - The main user, group, expense and settlement payloads, along with their required fields and decimal precision for money value, are covered by schema validation. - The service tests and repository tests validate the principal creation flows, transaction safety, and group membership behavior within an isolated SQLite test database. - The dashboard currently calculates the summary values on the client side from the expense history response. This is fine for the MVP, and also shows us where Sprint 6 can add a summary endpoint if required.

3. GitLab Issues & Work Division

GitLab Issues and Work Division.

For the fifth Sprint, the team organized the issues primarily along four implementation areas to keep the remaining work narrow and testable.

- Authentication and persistence of login state including account creation, login validation, and page session persistence.
- In the expense flow, one can create a group initially by setting a group structure, assigning members and reusing existing group if relevant.
- The expense history and the closure of an API, that is, what information goes into the list/create expense endpoints and the response shape which the dashboard and history pages use.
- Develop and execute settlement regression coverage, i.e. settlement persistence, activity logs, service/repo tests.

With this division, the sprint remains focused on the highest-value MVP paths. There is also a risk of scope creep into features that are better suited for Sprint 6.

4. Project Artifacts Updates

- The backend structure keeps its organization into models, schemas, repositories, services, and routers, and Sprint 5 work further solidified the relationships between those layers.
- The frontend pages of login, register, dashboard, add expense, and history matched current API behavior and shared login state.
- The engineers broadened the automated test suite around repository and service behaviour, focusing on group creation, expense splits and settlement persistence.
- Currently the database schema and ORM layer represents the main MVP entities as users, sessions, password reset tokens, groups, group members, expenses, expense participants, settlements and activity log.

5. Weekly Meeting Summary

Weekly meeting summary.

During the weekly meeting, we reviewed the current authentication flow, our approach for login-state persistence, plus what is left to make group and settlement flow feel complete. We also discussed the present trade-off of using client-side dashboard aggregation from `/api/expenses` rather than creating a summary endpoint immediately.

The team decided that the focus of Sprint 5 should remain on closing the core user journey first: sign in, keep the user context available across pages, create or reuse a group, add shared expense, expose the resulting data in stable format. As a result, we were able to realistically see a pathway towards a smaller but more reliable Sprint 6.

6. Challenges & Learnings

- One challenge was managing to keep the log-in state logic simple, without pretending it has a full server-side session. By using saved user data on frontend and having clear auth flow on backend, we handled this.
- The boundaries of workflows between groups also posed a challenge. The repository and service logic already have group membership covered well. However, not every group action is yet exposed as its own route.

- Using expense history endpoint as the source of the dashboard data is a reasonable MVP decision, but it will make our dashboard responsible for more client-side aggregation than we would want in a later production version.
- The repository pattern has proven useful in this project's test suite since it allows us to test the behaviour of transactions without relying on an external environment.

7. Next Steps (Sprint 6 Plan)

7. What to do next (Plan for Sprint 6).

The remaining MVP features that rely on the existing login, group, and expense base will be targeted in Sprint 6

- Settlement flow: display and link the remaining settlement actions to make it easier for users to register repayments and see balance changes.
- Add a simple mechanism for reminding me of my dues remaining to be paid or shares not transferred.
- The only administrator functions which are essential to the course scope should be implemented (minimum useful version). Clearly delineated boundaries on permissions should be indicated.
- Carry out a full run on the authentication and group creation, expense creation, expense history and settlement flows to catch integration breaks before the final submission.

In Sprint 6, we plan to complete the practical MVP surface area and ensure a stable codebase while ensuring that the final demo path is predictable end to end.